

Runna

Traceability Record

Wanikkasit Nopthiraitthikun 652115043

Thanachai Naksomboon 662115020

Bachelor of Science

Software Engineering Program

College of Arts, Media, and Technology

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Project Advisor

Assistant Professor Dr. Noppon Choosri

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Chapter 1 | Introduction

1.1 Purpose

The purpose of this Traceability Record is to establish and maintain traceability between the system requirements, design components, and testing activities of the Runna application. This document ensures that every functional requirement defined for the system is properly addressed during system design and verified through corresponding test cases.

The traceability process helps confirm that all implemented features are aligned with project objectives and that no requirement is omitted during development and testing. This document also supports project monitoring, quality assurance, and future maintenance activities.

1.2 Scope

This Traceability Record covers all major features implemented in Progress 1 of the Runna project, including:

- User Authentication
- Route Management
- GPS Run Tracking
- Running Statistics Calculation
- AI-Based Running Analysis
- Hazard Pin Management
- Administrative Functions

The document maps each requirement to its related design components and testing records to verify that the system satisfies the specified requirements.

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	Pct of task complete (%)	
	Progress I	Progress II
Feature 1: Authentication & Roles	100 %	100 %
1.3 Register	✓	✓
1.4 Login	✓	✓
1.5 Logout	✓	✓
1.6 Password Reset	✓	✓
1.7 Role-Based Access Control	✓	✓
Feature 2: Route Creation & Tracking	50%	100 %
1.1 View Map	✓	✓
1.2 Custom Route Creation	✓	✓
1.3 Route Editing(Add/Remove Waypoints)	✓	✓
1.4 Statistic Calculation	✗	✓
1.5 Real-time Tracking (Start/Stop Run)	✗	✓
1.6 Run History Storage	✗	✓
Feature 3: Collaborative Pin System	60%	100 %
1.3 Create Pin	✓	✓
1.4 View Pin on Map	✓	✓
1.5 Pin Categories	✓	✓
1.6 User Confirmation	✗	✓
1.7 Pin Status Management(Active/Expired)	✗	✓
Feature 4: AI Run Summary	65%	100%
4.1 Send Data to Gemini API	✓	✓
4.2 AI Analysis & Insight Generation	✓	✓
4.3 Recommendation Generation	✗	✓
Feature 5: Admin Dashboard	35%	100%
5.1 Manage users	✓	✓
5.2 Remove incorrect hazard reports	✗	✓
5.3 Manage route	✗	✓

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1.3 Acronyms

MD: Method Description

SRS: System Requirement Specification

STC: System Test Case

UC: Use Case Diagram

UD: Use Case Description

URS: User Requirement Specification

UTC: Unit Test Case

UI: User Interface

AI: Artificial Intelligence

API: Application Programming Interface

GPS: Global Positioning System

UI: User Interface

JWT: JSON Web Token

DB: database

ERD: Entity Relationship Diagram

FR: Functional Requirement

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Chapter 2 | Traceability Record Table

Feature #1 Authentication

Use Case	Use Case Description	User Requirement	System Requirement	Method Description	User Interface	Unit Test Case	System Test Case
UC-01 User Registration	User creates a new account	Users can create an account	System shall store user information securely	Register user account	Registration Screen	UTC-01	STC-01
UC-02 User Login	User logs into the system	Users can access their account	System shall validate credentials	Authenticate user and generate JWT token	Login Screen	UTC-01	STC-01
UC-03 User Logout	User logs out from the system	Users can safely exit the application	System shall terminate active session	Logout user session	Profile Screen	UTC-01	STC-01

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Feature #2 Route Management

Use Case	Use Case Description	User Requirement	System Requirement	Method Description	User Interface	Unit Test Case	System Test Case
UC-04 Create Route	User creates a route by selecting points on the map	Users can create personalized routes	System shall generate a route from selected coordinates	Create manual route	Route Creation Screen	UTC-02	STC-02
UC-05 View Route	User views created route	Users can review saved routes	System shall retrieve route information	Display route on map	Map Screen	UTC-02	STC-02
UC-06 Save Route	User saves route	Users can reuse routes later	System shall store route information	Save route data	Route Screen	UTC-02	STC-02

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Feature #3 Run Tracking & Statistics

Use Case	Use Case Description	User Requirement	System Requirement	Method Description	User Interface	Unit Test Case	System Test Case
UC-07 Start Run	User starts a running session	Users can begin tracking activity	System shall create a new run session	Start run	Running Screen	UTC-04	STC-02
UC-08 Track Location	User movement is tracked during running	Users can monitor running progress	System shall record GPS points	Record GPS coordinates	Running Screen	UTC-03	STC-02
UC-09 Stop Run	User finishes running	Users can complete activity recording	System shall close running session	Finish run	Running Screen	UTC-04	STC-02
UC-10 View Statistics	User reviews running results	Users can view performance metrics	System shall calculate distance, duration, pace and steps	Generate running statistics	Summary Screen	UTC-04	STC-02
UC-11 View Run History	User reviews previous runs	Users can access past activities	System shall store and retrieve completed runs	Retrieve run history	History Screen	UTC-04	STC-02

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Feature #4 AI Analysis

Use Case	Use Case Description	User Requirement	System Requirement	Method Description	User Interface	Unit Test Case	System Test Case
UC-12 Generate Analysis	User receives running analysis	Users can understand their performance	System shall generate AI insights	Analyze running statistics	Analysis Screen	UTC-05	STC-02
UC-13 Generate Recommendation	User receives improvement suggestions	Users can improve future runs	System shall provide recommendations	Generate recommendations	Analysis Screen	UTC-05	STC-02
UC-14 Fallback Analysis	Analysis is generated when AI service fails	Users can still receive feedback	System shall use rule-based analysis	Generate fallback insights	Analysis Screen	UTC-05	STC-02

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Feature #5 Hazard Pin Management

Use Case	Use Case Description	User Requirement	System Requirement	Method Description	User Interface	Unit Test Case	System Test Case
UC-15 Create Hazard Pin	User reports a hazard location	Users can report dangerous areas	System shall store hazard reports	Create hazard pin	Hazard Pin Screen	UTC-06	STC-03
UC-16 View Hazard Pins	User views hazards on map	Users can identify unsafe locations	System shall display hazard markers	Display hazard pins	Map Screen	UTC-06	STC-03
UC-17 View Hazard Details	User views hazard information	Users can access hazard descriptions	System shall retrieve hazard information	Display hazard details	Hazard Detail Screen	UTC-06	STC-03

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